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### Refrigeration system having a heat pump function for a motor vehicle, having a single sensor device on the low-pressure side

#### Abstract

A refrigeration system having a heat pump function for a motor vehicle. The refrigeration system includes: a refrigerant compressor which is connectable or connected to a primary line; a directly or indirectly acting external heat exchanger, which is arranged in the primary line; a first evaporator, which is arranged in the primary line; a first directly or indirectly acting heat exchanger, in particular a chiller, which is arranged fluidically in parallel to the evaporator; and a refrigerant collector arranged on the low-pressure side. A single sensor device is arranged downstream of the evaporator and the further heat exchanger, in particular the chiller, which is configured to detect the pressure and the temperature of the refrigerant on the low-pressure side of the refrigeration system.

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|------------------------------|----------------------------------------------------------------------------------------------------------|
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## Background/Summary

### FIELD

(1) The invention relates to a refrigeration system, in particular having a heat pump function, for a motor vehicle, wherein the refrigeration system comprises: a refrigerant compressor which is connectable or connected to a primary line; a directly or indirectly acting external heat exchanger, which is arranged in the primary line; a first evaporator, which is arranged in the primary line; a first directly or indirectly acting heat exchanger, in particular a chiller, which is arranged fluidically in parallel to the evaporator; and a refrigerant collector arranged on the low-pressure side.

### BACKGROUND

(2) According to DE 10 2018 201 165 B3, refrigeration systems having a refrigerant collector on the low-pressure side, which use both direct waste heat from electrical (drive) components via the additional heat exchanger, in particular a chiller, and ambient heat via an external heat exchanger, are provided with pressure/temperature sensors at various points in the refrigerant circuit. In particular, two or more sensor devices for detecting pressure and/or temperature are regularly provided on the low-pressure side of the refrigeration system. Further refrigeration systems are known from DE 10 2015 007 564 A1, subsequently published DE 10 2019 135 056, DE 10 2018 201 945 A1, and DE 10 2018 213 232 A1. In this case, in particular on the low-pressure side, a pressure/temperature sensor is arranged in each case after the evaporator or after the additional heat exchanger (chiller) and after the refrigerant collector.

### SUMMARY

(3) The object on which the invention is based is considered that of specifying a refrigeration system in which a simplified structure is achieved with essentially the same functionality.

(4) A refrigeration system is thus proposed, in particular having an optional heat pump function, for a motor vehicle, wherein the refrigeration system comprises: a refrigerant compressor which is connectable or connected to a primary line; a directly or indirectly acting external heat exchanger, which is arranged in the primary line; a first evaporator, which is arranged in the primary line; a first directly or indirectly acting heat exchanger, in particular a chiller, which is arranged fluidically in parallel to the evaporator; and a refrigerant collector arranged on the low-pressure side. It is provided that a single sensor device is arranged downstream of the evaporator and the further heat exchanger, in particular the chiller, which is configured to detect the pressure and the temperature

of the refrigerant on the low-pressure side of the refrigeration system.

(5) By arranging a single sensor device for detecting pressure and temperature, one sensor device can be dispensed with in comparison to the known refrigeration systems, which simplifies the structure of the refrigeration system. It is to be noted that the single sensor device can have a single pressure sensor and a single temperature sensor or can be designed as a combined pressure/temperature sensor.

(6) In the refrigeration system, the single sensor device can be arranged between the refrigerant collector and the refrigerant compressor. In other words, the single sensor device can be arranged downstream of the refrigerant collector. Compared to the known refrigeration systems, the position of the single sensor device after the refrigerant collector is retained, wherein the pressure/temperature sensor after the evaporator or chiller is simply dispensed with.

(7) Alternatively, the single sensor device can be arranged upstream of the refrigerant collector in the refrigeration system. In other words, the single sensor device is arranged before the refrigerant collector, in particular shortly or directly before the refrigerant collector. In this context, immediately before the refrigerant collector can be understood to mean a distance that extends up to approximately 20 centimeters upstream of the refrigerant collector.

(8) In both alternatives, the single sensor device can be arranged downstream of a branch in which an outlet-side line section of the evaporator and an outlet-side line section of the further heat exchanger, in particular a chiller, are connected to one another. In other words, the single sensor device is arranged after the refrigerant lines of the evaporator and the chiller have been combined.

(9) The refrigeration system can have a control device that is configured to carry out at least one of the following functions based on detected pressure and/or temperature values from the single sensor device: monitoring temporary overheating of the refrigerant, in particular occurring for a short or long time, to identify underfilling; monitoring the minimum suction pressure for anti-icing protection. Brief overheating with accompanying underfilling, which usually lasts only a few seconds, is generally not critical for the operation of the refrigeration system, whereas long-term overheating with accompanying underfilling of at least one to several minutes is disadvantageous and critical for the operation of the refrigeration system.

(10) An overheating condition in particular can be taken into consideration in the monitoring and, if necessary, timely compensation or the subsiding of the temporary underfilling. Such an overheating condition can occur, for example, in the event of load changes in the system towards a higher power requirement; low load when the evaporator is operated alone in cooling operation (AC operation); a heat pump operation, or in the event of a post-heating operation (reheat) with excess heat.

(11) With the arrangement of the single sensor device downstream of the refrigerant collector, due to the transition from stationary to at least temporarily dynamic (transient) operation of the refrigeration system in connection with the load change that has occurred and the release of additionally required refrigerant from the refrigerant collector that has not yet completely taken place into the circuit, a delay in relieving the overheating condition during regular system operation is to be taken into consideration. The process of supplying refrigerant from the collector into the active system or the circuit itself runs with a delay. The larger the load change, the more pronounced or longer is the period of time before regular, quasi-stationary operation with an essentially constant refrigerant mass flow is restored.

(12) In the refrigeration system having the single sensor device on the low-pressure side upstream of the refrigerant collector, the control device can also be configured to carry out at least one of the following additional functions based on detected pressure and/or temperature values: regulating the evaporator and/or the additional heat exchanger, in particular a chiller, in an AC dual operation; regulating the overheating of the evaporator if more cooling power is required at the additional heat exchanger, in particular the chiller; regulating the overheating of the additional heat exchanger, in particular the chiller, if more cooling power is required at the evaporator.

- (13) If the single sensor device is arranged upstream of the refrigerant collector, the above-described functions with respect to overheating can be implemented without any significant delay. Accordingly, the monitoring and, if necessary, curtailment of temporary underfilling via the overheating condition can take place very quickly and precisely.
- (14) In the refrigeration system having the single sensor device on the low-pressure side upstream of the refrigerant collector, the control device can be configured to take into consideration at least one stored characteristic curve and/or at least one stored characteristic diagram, wherein the characteristic curve or the characteristic diagram comprises values for determining prevailing pressure positions, in particular a suction pressure curtailment, based on a pressure detected by the single sensor device and on a pressure loss contained in the characteristic curve or in the characteristic diagram, which occurs downstream of the single sensor device to the refrigerant compressor. The characteristic curve or the characteristic diagram is used in particular to achieve improved quality with regard to the determination or estimation of target variables that are set by the system. Efforts are taken here so that the required or current pressures are determined at all operating points of the refrigeration system and, if necessary, a low-pressure limit is estimated or predicted.
- (15) Furthermore, in the refrigeration system, the control device can be configured to recognize underfilling, in particular by means of the evaluated refrigerant condition, downstream of the refrigerant collector, wherein a delaying effect of the refrigerant collector is estimated or determined by means of a characteristic diagram and/or by means of functions. It is to be taken into consideration that recognizing underfilling is detected more directly and therefore more dynamically and possibly more sensitively using the sensor device arranged upstream of the refrigerant collector, so that the damping or delaying effect described due to the refrigerant collector is to be modeled to set a final refrigerant state in the refrigeration system by means of a characteristic diagram and/or functions. In a simplified variant, a characteristic curve can alternatively be used.
- (16) The refrigeration system can furthermore comprise: a secondary line connected or connectable to the refrigerant compressor; and another heat exchanger that represents a heat source and acts directly or indirectly, in particular a heating register, which is arranged in the secondary line.
- (17) The refrigeration system can furthermore comprise: a primary line valve arranged between the refrigerant compressor and the external heat exchanger; and a secondary line valve arranged between the refrigerant compressor and the additional heat exchanger, in particular a heating register, which represents a heat source.
- (18) A motor vehicle, in particular an at least partially electrically operated motor vehicle, can have an above-described refrigeration system. In an electric vehicle, the efficient operation of the refrigeration system can result in power savings, so that a greater range of the electric vehicle can be achieved as a result. In particular, essentially all operating states of the refrigeration system can also be monitored by the refrigeration system presented here using only a single sensor device on the low-pressure side for detecting pressure and temperature, as if the refrigeration system were to have at least two pressure/temperature sensors on the low-pressure side.

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## Description

### BRIEF DESCRIPTION OF THE FIGURES

- (1) Further advantages and details of the invention result from the following description of embodiments with reference to the figures. In the figures:
- (2) FIG. 1 shows a schematic and simplified circuit diagram of a refrigeration system for a motor vehicle;
- (3) FIG. 2 shows a schematic and simplified circuit diagram of a refrigeration system for a motor

vehicle;

## DETAILED DESCRIPTION

(4) FIG. 1 shows an embodiment of a refrigeration system **10** for a motor vehicle in a schematic and simplified manner. The refrigeration system **10** comprises a refrigerant circuit **11**, which can be operated both in a refrigeration system operation (also called AC operation for short) and in a heat pump mode. In the embodiment shown, the refrigeration system **10** comprises a refrigerant compressor **12**, an external heat exchanger **18**, an internal heat exchanger **20**, an evaporator **22** and an accumulator or refrigerant collector **24**. The external heat exchanger **18** can be designed as a condenser or gas cooler. In particular, the external heat exchanger **18** can have flow through it in both directions in the illustrated embodiment.

(5) The evaporator **22** is shown here by way of example as a front evaporator for a vehicle. The evaporator **22** is also representative of other evaporators possible in a vehicle, for example rear evaporators, which can be arranged fluidically in parallel to one another. In other words, the refrigeration system **10** comprises at least one evaporator **22**.

(6) A shut-off valve **A4** is arranged downstream of the compressor **12**. An expansion valve **AE2** is provided upstream of the evaporator **22**.

(7) In the scope of this description, the section from the compressor **12** to the outer heat exchanger **18**, to the inner heat exchanger **20**, and to the evaporator **22** is referred to as the primary line **14** in the entire refrigerant circuit **11** of the refrigeration system **10**.

(8) The refrigeration system **10** furthermore comprises a heating register **26** (also referred to as a heating condenser or heating gas cooler). A shut-off valve **A3** is arranged upstream of the heating register **26**. A shut-off valve **A1** is arranged downstream of the heating register **26**. Furthermore, an expansion valve **AE4** is arranged downstream of the heating register **26**.

(9) In the scope of this description, the section from the compressor **12** to the heating register **26**, to the expansion valve **AE4** and to a branch **Ab2** is referred to as the secondary line **16** in the entire refrigerant circuit of the refrigeration system **10**. The secondary line **16** comprises a heating branch **16.1**, which extends from the shut-off valve **A3** via the heating register **26** to the shut-off valve **A1**. The secondary line **16** also includes a reheating branch or reheat branch **16.2**, which is fluidically connectable to the heating register **26** upstream and to the external heat exchanger **18** downstream. The secondary line **16** or the reheat branch **16.2** opens into the primary line **14** at a branching point **Ab2**.

(10) The refrigeration system **10** comprises a further evaporator or chiller **28**. The chiller **28** is provided fluidically in parallel to the evaporator **22**. The chiller **28** can be used, for example, to cool an electrical component of the vehicle, but also to implement a water heat pump function using the waste heat from at least one electrical component. An expansion valve **AE1** is connected upstream of the chiller **28**.

(11) The refrigeration system **10** can also have an electrical heating element **30**, which is designed, for example, as a high-voltage PTC heating element. The electric heating element **30** is used as an auxiliary heater for a supply air flow **L** guided into the vehicle interior. The electric heating element **30** can be accommodated in an air conditioning unit **32** together with the heating register **26** and the evaporator **22**. In this case, the electrical heating element **30** can be arranged downstream of the heating register **26**.

(12) Check valves **R1** and **R2** are also apparent in FIG. 1. Furthermore, several sensors **pT1**, **pT5**, **pT6** on the high-pressure side for detecting pressure and/or temperature of the refrigerant are also shown. It is to be noted that the number of sensors on the high-pressure side and their arrangement is only shown here as an example. In the example shown, combined pressure/temperature sensors **pT1**, **pT5**, and **pT6** are shown as the sensors. However, it is just as conceivable that sensors that are separate from one another are used for measuring pressure or temperature and, if necessary, are also arranged spatially separately from one another along the refrigerant lines.

(13) The refrigeration system **10** can be operated in different modes, which are briefly described

below.

(14) In AC operation of the refrigerant circuit **11**, the refrigerant compressed to high pressure flows from the refrigerant compressor **12** when the shut-off valve **A4** is open into the outer heat exchanger **18**. From there it flows to the high-pressure section of the internal heat exchanger **20** and the fully open expansion valve **AE3**. The refrigerant can flow to the expansion valve **AE2** and into the interior evaporator **22** via a branching point **Ab1** (evaporator section **22.1**). In parallel or alternatively, the refrigerant can flow into the chiller **28** (chiller section **28.1**) via a branching point **Ab4** and the expansion valve **AE1**. From the evaporator **22** and/or the chiller **28**, the refrigerant flows on the low-pressure side into the collector **24** and through the low-pressure section of the internal heat exchanger **20** back to the compressor **12**.

(15) In AC operation, the heating branch **16.1** or the secondary line **16** is shut off by means of the shut-off valve **A3**, so that hot refrigerant cannot flow through the heating register **26**. To retrieve refrigerant from the inactive heating branch **16.1**, the shut-off element **A5**, which is designed as a shut-off valve, can be opened so that the refrigerant can flow in the direction of the collector **24** via the shut-off element **A5** and the check valve **R2**, with the shut-off element **A2** being closed at the same time.

(16) In heating operation of the refrigerant circuit **11**, the shut-off valve **A4** is closed and the shut-off valve **A3** is open, so that hot refrigerant can flow into the heating branch **16.1**.

(17) To carry out the heating function by means of the chiller **28** to implement water heat pump operation, the refrigerant compressed by means of the refrigerant compressor **12** flows into the heating register **26** via the open shut-off valve **A3**. At the heating register **26**, heat is given off to a supply air flow **L** guided into the vehicle interior. The refrigerant then flows via the open shut-off valve **A1** and the branching point **Ab1**. It is expanded by means of the expansion valve **AE1** in the chiller **28** to absorb waste heat from electrical and/or electronic components arranged in a coolant circuit **28.2**. With this heating function, the expansion valves **AE3** and **AE4** are closed, the shut-off valve **A5** is closed, and the shut-off valve **A2** is open. In this case, refrigerant displaced in water heat pump operation can be extracted via the shut-off valve **A2** out of a bidirectional line **14.1** or the primary line **14** and supplied to the collector **24** via the check valve **R2**.

(18) To carry out the heating function by means of the external heat exchanger **18** as a heat pump evaporator, the refrigerant compressed by means of the refrigerant compressor **12** flows via the open shut-off valve **A3** to give off heat to a supply air flow **L** into the heating register **26**. It is then expanded via the open shut-off valve **A1** by means of the expansion valve **AE3** in the outer heat exchanger **18** to absorb heat from the ambient air. The refrigerant then flows via a heat pump return branch **15** to the collector **24** and back to the refrigerant compressor **12**. The expansion valves **AE1**, **AE2**, and **AE4** remain closed, as does the shut-off valve **A5**.

(19) An indirect delta connection can be implemented in that when the shut-off valve **A1** is open, the refrigerant compressed by the refrigerant compressor **12** is expanded by means of the expansion valve **AE1** in the chiller **28**, wherein no mass flow is generated at the same time on the coolant side, i.e., in the coolant circuit **28.2**, thus, for example, the fluid used as the coolant, such as water or water-glycol mixture, remains on the coolant side of the chiller **28** or coolant does not actively flow through the chiller **28**. The expansion valves **AE2**, **AE3**, and **AE4** remain closed in this switching variant.

(20) In a reheating or reheat operation, the supply air flow **L** supplied into the vehicle interior is first cooled by means of the evaporator **22** and thus dehumidified. Using the heat transferred to the refrigerant by evaporation and dehumidification and the heat supplied to the refrigerant via the compressor **12**, the supply air flow **L** can be completely or at least partially reheated by means of the heating register **26**.

(21) For this purpose, the refrigeration system **10**, in particular the air conditioning unit **32**, has adjustable, in particular controllable and pivotable, temperature flaps **34** between the evaporator **22** and the heating register **26**. In the example shown, a left and a right temperature flap **34L** and **34R**

(shown schematically in FIG. 1) are arranged. The temperature flaps **34L**, **34R** can be adjusted or pivoted between an open position, designated as a 100% position, and a closed position, designated as a 0% position. Alternatively, it is also possible to connect the temperature flaps **34R**, **34L** downstream of the heating register **26**.

(22) In the 100% position, the entire supply air flow **L** flowing through the evaporator **22** is guided and heated via the heating register **26** before it can flow into the passenger compartment of the vehicle. In the 0% position the entire supply air flow **L** flowing through the evaporator **22** flows in the bypass around the heating register **26** without heating and thus without absorbing heat into the passenger compartment.

(23) In an  $x$  position of the temperature flaps **34L** and **34R** with  $0\% < x < 100\%$ , these temperature flaps are only partially open, so that in each case only a partial air flow of the supply air flow **L** flowing through the evaporator **22** is guided via the heating register **26**. This heated partial air flow can then be mixed with the remaining, cooled and dehumidified partial air flow. The supply air flow **L** heated in this way is supplied to the passenger compartment of the vehicle. As an example, a 50% position indicates that the temperature flaps **34R** and **34L** are only half open, i.e., 50%.

(24) A single sensor device **pT2** is arranged on the low-pressure side of the refrigeration system **10**, in particular downstream of the evaporator **22** and downstream of the chiller **28**. In the example in FIG. 1, the sensor device **pT2** is arranged downstream of the refrigerant collector **24**. In particular, the sensor device **pT2** is provided between the refrigerant collector **24** and the internal heat exchanger **20**.

(25) The evaporator **22** and the chiller **28** have a respective section, arranged on the outlet side or downstream of the relevant evaporator section **22.1** or chiller section **28.1**, respectively, which are connected to one another at the branch **Ab2**. The single sensor device **pT2** on the low-pressure side is provided downstream of the branch **Ab2**. This ensures that the pressure and the temperature of the refrigerant can be detected on the low-pressure side, independently of whether refrigerant flows through the evaporator **22** or the chiller **28** alone or in combination. In other words, the single sensor device **pT2** is arranged between the branch **Ab2** and the internal heat exchanger **20** (if present) or the refrigerant compressor **12**.

(26) FIG. 2 shows an alternative arrangement of the single low-pressure side sensor device **pT2**. In this case, the sensor device **pT2** is arranged upstream of the refrigerant collector **24**. In particular, the sensor device **pT2** can be arranged directly before the refrigerant collector **24**. A distance between the sensor device **pT2** and the refrigerant collector can be up to 20 centimeters, for example.

(27) The sensor device **pT2** is also arranged downstream of the evaporator **22** and the chiller **28** in the example in FIG. 2. In particular, the sensor device **pT2** is arranged downstream of the branch **Ab2**, where the evaporator section **22.1** and the chiller section **28.1** meet. In other words, the sensor device **pT2** is arranged between the branch **Ab2** and the coolant collector **24**.

(28) In both examples of FIGS. 1 and 2, the refrigeration system **10** has a control device **50** which is configured to detect the various operating states of the refrigeration system **10** that have been described above by way of example, but in particular to control or regulate them on the basis of the input values and setpoint variables. The control device **50** can in particular receive and process sensor data, calculate and/or estimate further operating parameters, and activate the refrigerant compressor or valve devices and the like of the refrigeration system.

(29) In consideration of the structural design having a single low-pressure-side sensor device **pT2**, the control device **50** can be configured to perform at least one of the following functions based on recorded pressure and/or temperature values from the single low-pressure-side sensor device **pT2**: monitoring overheating, equivalent to systems having a refrigerant collector **24** arranged on the low-pressure side with underfilling, which in turn can occur either temporarily or permanently; and/or monitoring the minimum suction pressure for anti-icing protection.

(30) In particular, an underfill condition can be taken into consideration when monitoring and, if

necessary, curtailing the temporary overheating. Such an overheating and thus underfilling condition can occur, for example, in the event of load changes in the system towards a higher power requirement; low load when the evaporator is operated alone in cooling operation (AC operation); a heat pump operation, or a post-heating operation (reheat) with excess heat.

(31) With the arrangement of the single sensor device pT2 downstream of the refrigerant collector **24** (FIG. 1), due to the transition from stationary to at least temporarily dynamic (transient) operation of the refrigeration system **10** in connection with the load change that has occurred and the release of additionally required refrigerant from the refrigerant collector **24** that has not yet completely taken place into the circuit, a delay in relieving the overheating condition during regular system operation is to be taken into consideration. The process of supplying refrigerant from the collector **24** into the active system or the circuit **11** itself runs with a delay. The larger the load change, the more pronounced or longer is the period of time before regular, quasi-stationary operation with an essentially constant refrigerant mass flow is restored.

(32) In the refrigeration system **10** having the low pressure-side sensor device pT2 upstream of the refrigerant collector **24** (FIG. 2), the control device **50** can also be configured to carry out at least one of the following additional functions based on detected pressure and/or temperature values: regulating the evaporator **22** and/or the additional heat exchanger, in particular a chiller **28**, in an AC dual operation; regulating the overheating of the evaporator **22** if more cooling power is required at the additional heat exchanger, in particular the chiller **28**; regulating the overheating of the additional heat exchanger, in particular the chiller **28**, if more cooling power is required at the evaporator **22**.

(33) If the single sensor device pT2 is arranged upstream of the refrigerant collector **24** (FIG. 2), the above-described functions with respect to overheating can be implemented without any significant delay. Accordingly, the monitoring and, if necessary, curtailment of temporary underfilling via the overheating state can take place very quickly and precisely. In the extreme case, which corresponds, for example, to permanently occurring overheating, which is measured, for example, at greater than 5 K, and unsuccessful regulator intervention to eliminate this normal operating deviation, the result is that the refrigeration system **10** is switched off.

(34) In the refrigeration system **10** having the single low-pressure-side sensor device pT2 upstream of the refrigerant collector **24** (FIG. 2), the control device **50** can be configured to take into consideration at least one stored characteristic diagram, wherein the characteristic diagram comprises values for determining a suction pressure regulation based on a pressure detected by the single sensor device and on a pressure loss contained in the characteristic diagram, which occurs downstream from the single sensor device pT2 to the refrigerant compressor **12**.

(35) Furthermore, in the refrigeration system **10**, the control device **50** can be configured to recognize underfilling downstream of the refrigerant collector **24**, wherein a delaying effect of the refrigerant collector **24** is estimated by means of a characteristic diagram and/or by means of functions. It is to be taken into consideration that recognizing underfilling is detected more directly and therefore more dynamically and possibly more sensitively by the sensor device pT2 (FIG. 2) arranged upstream of the refrigerant collector **24**, so that the damping or delaying effect described due to the refrigerant collector **24** arranged downstream is to be modeled by means of a characteristic diagram and/or functions.

(36) By means of the refrigeration system **10** presented here having a single low-pressure-side sensor device pT2, at least one pressure-temperature sensor can be saved compared to known configurations of refrigeration systems for motor vehicles, which has a positive effect on the costs, package requirements, and weight of such a refrigeration system. Furthermore, despite the use of a single low-pressure-side sensor device pT2, it is possible to represent all essential functions or to provide them by means of the control device **50** of the refrigeration system **10**.

## Claims

1. A refrigeration system having heat pump function for a motor vehicle, comprising: a refrigerant compressor, which is connectable or connected to a primary line; a directly or indirectly acting external heat exchanger, which is arranged in the primary line; a first evaporator, which is arranged in the primary line; a first directly or indirectly acting heat exchanger, comprising a chiller, which is arranged fluidically parallel to the first evaporator; a refrigerant collector arranged on the low-pressure side, and a control device, wherein a single sensor device is arranged downstream of the first evaporator and the first directly or indirectly acting heat exchanger comprising the chiller, wherein the single sensor device is configured to detect the pressure and the temperature of the refrigerant on the low-pressure side of the refrigeration system, and wherein the control device is configured to obtain one or more detected pressure and/or temperature values from the single sensor device and is configured to perform one of: monitoring temporary, in particular short-term or long-term, overheating of the refrigerant for recognizing underfilling; and monitoring the minimum suction pressure for anti-icing protection.
2. The refrigeration system as claimed in claim 1, wherein the single sensor device is arranged between the refrigerant collector and the refrigerant compressor.
3. The refrigeration system as claimed in claim 1, wherein the single sensor device is arranged upstream of the refrigerant collector.
4. The refrigeration system as claimed in claim 2, wherein the single sensor device is arranged downstream of a branch, in which an outlet-side line section of the first evaporator and an outlet-side line section the first directly or indirectly acting heat exchanger are connected to one another.
5. The refrigeration system as claimed in claim 1, wherein the control device is configured to carry out at least one of the following further functions based on detected pressure and/or temperature values from the single sensor device arranged upstream of the refrigerant collector: regulating the first evaporator and/or the first directly or indirectly acting heat exchanger in an AC dual operation; regulating the overheating of the first evaporator if more cooling power is required at the first directly or indirectly acting heat exchanger; regulating the overheating of the first directly or indirectly acting heat exchanger if more cooling power is required at the evaporator.
6. The refrigeration system as claimed in claim 5, wherein the control device is configured to take input from at least one stored characteristic curve and/or at least one stored characteristic diagram, wherein the at least one stored characteristic curve or the at least one stored characteristic diagram comprises values for determining prevailing pressure levels, in particular a suction pressure curtailment, based on a pressure detected by the single sensor device and a pressure loss contained in the characteristic curve or in the characteristic diagram, which occurs downstream from the single sensor device to the refrigerant compressor.
7. The refrigeration system as claimed in claim 6, wherein the control device is furthermore configured to recognize underfilling downstream of the refrigerant collector, wherein a delaying effect of the refrigerant collector is estimated or determined based on at least one of: a characteristic diagram the monitoring functions.
8. The refrigeration system as claimed in claim 1, further comprising: a secondary line, which is connectable or connected to the refrigerant compressor; and a further directly or indirectly acting heat exchanger, representing a heat source, comprising a heating register, which is arranged in the secondary line.
9. The refrigeration system as claimed in claim 8, further comprising: a primary line valve arranged between the refrigerant compressor and the directly or indirectly acting external heat exchanger; a secondary line valve arranged between the refrigerant compressor and the further directly or indirectly acting heat exchanger.
10. A motor vehicle, comprising an at least partially electrically powered motor vehicle, having the

refrigeration system as claimed in claim 1.

11. The refrigeration system as claimed in claim 3, wherein the single sensor device is arranged downstream of a branch, in which an outlet-side line section of the first evaporator and an outlet-side line section of the first directly or indirectly acting heat exchanger, are connected to one another.

12. The refrigeration system as claimed in claim 2, further comprising: a secondary line, which is connectable or connected to the refrigerant compressor; and a further directly or indirectly acting heat exchanger, representing a heat source comprising a heating register, which is arranged in the secondary line.

13. The refrigeration system as claimed in claim 3, further comprising: a secondary line, which is connectable or connected to the refrigerant compressor; and a further directly or indirectly acting heat exchanger, representing a heat source, comprising a heating register, which is arranged in the secondary line.

14. The refrigeration system as claimed in claim 4, further comprising: a secondary line, which is connectable or connected to the refrigerant compressor; and a further directly or indirectly acting heat exchanger, representing a heat source, comprising a heating register, which is arranged in the secondary line.

15. The refrigeration system as claimed in claim 5, further comprising: a secondary line, which is connectable or connected to the refrigerant compressor; and a further directly or indirectly acting heat exchanger, representing a heat source, comprising a heating register, which is arranged in the secondary line.

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